Diag Solutions

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1. **1.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building is 45°. The height above his eye level is:

   Answer: (B)

   Solution:

   1. Height of building = 30 m.
   2. Height of boy = 1.5 m.
   3. Height above eye level = 30 − 1.5 = 28.5 m.
   4. Since angle is 45°, height = base distance.

   Derivation:

   $$h = 30 - 1.5 = 28.5 \text{ m}$$

   Some Applications of Trigonometry — Heights and Distances

2. **2.** The shadow of a tower standing on a level ground is found to be 40 m longer when the Sun's altitude is 30° than when it is 60°. The height of the tower is not required, but if we consider a simpler case where the shadow is $h\sqrt{3}$ when angle is 30°, what is the height h if the shadow is 30 m?

   Answer: (C)

   Solution:

   1. $\tan(30^\circ) = \frac{h}{30}$
   2. $\frac{1}{\sqrt{3}} = \frac{h}{30}$
   3. $h = \frac{30}{\sqrt{3}} = 10\sqrt{3}$

   Derivation:

   $$h = 30 \times \tan(30^\circ) = 30 \times \frac{1}{\sqrt{3}} = 10\sqrt{3}$$

   Some Applications of Trigonometry — Heights and Distances

3. **3.** Assertion (A): The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is 45°. The height of the tower is 30 m. Reason (R): In a right-angled triangle, $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$.

   Answer: (A)

   Solution:

   1. Use the tangent ratio: $\tan(45^\circ) = \frac{h}{30}$.
   2. Since $\tan(45^\circ) = 1$, we get $1 = \frac{h}{30}$.

3. Thus, $h = 30$ m.
4. The reason is the correct definition of the tangent ratio.

Derivation:

$$\begin{aligned} \tan(45^\circ) &= \frac{\text{height}}{30} \\ \implies 1 &= \frac{h}{30} \\ \implies h &= 30 \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

4. **4.** Assertion (A): If the length of the shadow of a vertical pole is equal to its height, then the angle of elevation of the Sun is 45°. Reason (R): The angle of elevation is the angle formed by the line of sight with the horizontal when the object is above the horizontal level.

   Answer: (A)

   Solution:

   1. Let height be h and shadow be s. Given $h = s$.
   2. $\tan(\theta) = \frac{h}{s} = \frac{h}{h} = 1$.
   3. $\theta = 45^\circ$.
   4. Reason is the standard definition of angle of elevation.

   Derivation:

$$\begin{aligned} \tan(\theta) &= \frac{\text{height}}{\text{shadow}} = 1 \\ \implies \theta &= 45^\circ \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

5. **5.** Assertion (A): The length of a kite string making an angle of 30° with the ground to reach a height of 50 m is 100 m. Reason (R): $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$.

   Answer: (A)

   Solution:

   1. Use the sine ratio: $\sin(30^\circ) = \frac{50}{L}$.
   2. Since $\sin(30^\circ) = 0.5$, we get $0.5 = \frac{50}{L}$.
   3. Thus, $L = 100$ m.
   4. Reason is the standard definition of the sine ratio.

   Derivation:

$$\begin{aligned} \sin(30^\circ) &= \frac{50}{L} \\ \implies \frac{1}{2} &= \frac{50}{L} \\ \implies L &= 100 \end{aligned}$$

6. **6.** Assertion (A): The angle of depression from a point 10 m above sea level to a ship at sea is $30°$. The distance of the ship from the base of the point is $10\sqrt{3}$ m. Reason (R): Angle of depression is equal to the angle of elevation by alternate interior angles.

   Answer: (A)

   Solution:

   1. The angle of depression equals the angle of elevation ($30°$).
   2. $\tan(30°) = \frac{10}{d}$.
   3. $\frac{1}{\sqrt{3}} = \frac{10}{d} \implies d = 10\sqrt{3}$.
   4. Reason explains why the triangle is formed correctly.

   Derivation:

$$\begin{aligned} \tan(30°) &= \frac{10}{d} \\ \implies d &= 10 \times \sqrt{3} = 10\sqrt{3} \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

7. **7.** Assertion (A): If a ladder 10 m long is placed against a wall such that it makes an angle of $60°$ with the ground, then the foot of the ladder is 5 m away from the wall. Reason (R): $\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$.

   Answer: (A)

   Solution:

   1. Use the cosine ratio: $\cos(60°) = \frac{\text{base}}{10}$.
   2. Since $\cos(60°) = 0.5$, we get $0.5 = \frac{\text{base}}{10}$.
   3. Thus, base = 5 m.
   4. Reason is the standard definition of the cosine ratio.

   Derivation:

$$\begin{aligned} \cos(60°) &= \frac{\text{base}}{10} \\ \implies \frac{1}{2} &= \frac{\text{base}}{10} \\ \implies \text{base} &= 5 \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

8. **8.** A ladder 15 m long makes an angle of $60°$ with the wall. Find the distance of the foot of the ladder from the wall.

   Answer:

$$7.5 \text{ m}$$

Solution:

1. Let L be the length of the ladder (15 m) and θ be the angle with the wall ($60°$). The distance of the foot from the wall is the side opposite to the angle of $60°$ in the right triangle.
2. Using $\sin(60°) = \frac{\text{opposite}}{\text{hypotenuse}}$, we have $\frac{\sqrt{3}}{2} = \frac{x}{15}$, leading to $x = 7.5\sqrt{3}$ m.

Derivation:

$$\begin{aligned} \sin(60°) &= \frac{x}{15} \\ \implies x = 15 \times \frac{\sqrt{3}}{2} &\approx 12.99 \text{ m} \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

9. **9.** The angle of elevation of the top of a tower from a point on the ground, which is 30 m away from the foot of the tower, is $30°$. Find the height of the tower.

Answer:

$$10\sqrt{3} \text{ m}$$

Solution:

1. Let h be the height of the tower and $d = 30$ m be the distance from the foot.
2. Use $\tan(30°) = \frac{h}{d}$ to solve for h.

Derivation:

$$\begin{aligned} \tan(30°) &= \frac{h}{30} \\ \implies \frac{1}{\sqrt{3}} &= \frac{h}{30} \\ \implies h = \frac{30}{\sqrt{3}} &= 10\sqrt{3} \text{ m} \end{aligned}$$

Some Applications of Trigonometry — Angles of Elevation

10. **10.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is $60°$. Find the length of the string.

Answer:

$$40\sqrt{3} \text{ m}$$

Solution:

1. The height $h = 60$ m is the opposite side and the string l is the hypotenuse.
2. Use $\sin(60°) = \frac{h}{l}$ to find l.

Derivation:

$$\sin(60^\circ) = \frac{60}{l}$$
$$\implies \frac{\sqrt{3}}{2} = \frac{60}{l}$$
$$\implies l = \frac{120}{\sqrt{3}} = 40\sqrt{3} \text{ m}$$

Some Applications of Trigonometry — Heights and Distances

11. **11.** From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60°. If the distance between the building and the tower is $7\sqrt{3}$ m, find the height of the tower.

Answer:

$$28 \text{ m}$$

Solution:

1. Let the tower height be $H = h + 7$, where h is the height above the building level.
2. Use $\tan(60^\circ) = \frac{h}{7\sqrt{3}}$ to find h, then add 7.

Derivation:

$$\tan(60^\circ) = \frac{h}{7\sqrt{3}}$$
$$\implies \sqrt{3} = \frac{h}{7\sqrt{3}}$$
$$\implies h = 21$$
$$\implies H = 21 + 7 = 28 \text{ m}$$

Some Applications of Trigonometry — Angle of Elevation

12. **12.** A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked.

Answer:

$$19\sqrt{3} \text{ m}$$

Solution:

1. The height of the building above eye level is $30 - 1.5 = 28.5$ m.
2. Calculate the horizontal distance for 30° and 60° using cot and find the difference.

Derivation:

$$d = 28.5(\cot 30^\circ - \cot 60^\circ) = 28.5(\sqrt{3} - \frac{1}{\sqrt{3}}) = 28.5(\frac{2}{\sqrt{3}}) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \text{ m}$$

13. 13. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height 6 m. At a point on the plane, the angle of elevation of the bottom and the top of the flagstaff are 30° and 60° respectively. Find the height of the tower.

Answer:

$$3m$$

Solution:

1. Let h be the height of the tower and x be the distance from the point to the tower base.
2. Use $\tan(30°) = h/x$ to express x in terms of h.
3. Use $\tan(60°) = (h + 6)/x$ and substitute x.
4. Solve the resulting linear equation for h.

Derivation:

$$\begin{aligned} \tan 30° &= \frac{h}{x} \\ \implies x &= h\sqrt{3} \\ \tan 60° &= \frac{h+6}{x} \\ \implies \sqrt{3} &= \frac{h+6}{h\sqrt{3}} \\ 3h &= h + 6 \\ \implies 2h &= 6 \\ \implies h &= 3 \end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

14. 14. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked towards the building.

Answer:

$$19\sqrt{3}m$$

Solution:

1. Calculate the height above eye level: $30 - 1.5 = 28.5$ m.
2. Use $\tan(30°)$ to find the initial distance from the building.
3. Use $\tan(60°)$ to find the final distance from the building.
4. Subtract the two distances to find the distance walked.

Derivation:

$$\text{Height } H = 28.5 \text{ m}$$
$$d_1 = \frac{H}{\tan 30^\circ} = 28.5\sqrt{3}$$
$$d_2 = \frac{H}{\tan 60^\circ} = \frac{28.5}{\sqrt{3}} = 9.5\sqrt{3}$$
$$\text{Distance} = 28.5\sqrt{3} - 9.5\sqrt{3} = 19\sqrt{3}$$

Some Applications of Trigonometry — Heights and Distances

15. **15.** The angle of elevation of the top of a lighthouse from a boat is 30°. If the boat moves 100 m towards the lighthouse, the angle of elevation becomes 45°. Find the height of the lighthouse.

Answer:

$$50(\sqrt{3}+1)m$$

Solution:

1. Let h be the height of the lighthouse and y be the distance after moving.
2. Set up $\tan(45^\circ) = h/y$ which means $h = y$.
3. Set up $\tan(30^\circ) = h/(y+100)$.
4. Solve for h by substituting $y = h$.

Derivation:

$$h = y$$
$$\frac{h}{h+100} = \tan 30^\circ = \frac{1}{\sqrt{3}}$$
$$h\sqrt{3} = h + 100$$
$$h(\sqrt{3}-1) = 100$$
$$h = \frac{100}{\sqrt{3}-1} = 50(\sqrt{3}+1)$$

Some Applications of Trigonometry — Heights and Distances

16. **16.** A kite is flying at a height of 60 m above the ground. The string attached to the kite is temporarily tied to a point on the ground. The inclination of the string with the ground is 60°. Find the length of the string, assuming that there is no slack in the string.

Answer:

$$40\sqrt{3}m$$

Solution:

1. Identify the scenario as a right-angled triangle where the string is the hypotenuse.
2. The height is the side opposite to the 60° angle.
3. Use the sine ratio: $\sin(60^\circ) = \text{Opposite}/\text{Hypotenuse}$.

Derivation:

$$\begin{aligned}\sin 60^\circ &= \frac{60}{L}\\ \frac{\sqrt{3}}{2} &= \frac{60}{L}\\ L = \frac{120}{\sqrt{3}} = \frac{120\sqrt{3}}{3} &= 40\sqrt{3}\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

17. 17. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

Answer:

$$20(\sqrt{3}-1)m$$

Solution:

1. Let h be the height of the tower and x be the distance from the point to the building.
2. Use $\tan(45^\circ) = 20/x$ to find x.
3. Use $\tan(60^\circ) = (20+h)/x$ to solve for h.

Derivation:

$$\begin{aligned}\tan 45^\circ &= \frac{20}{x}\\ \implies x &= 20\\ \tan 60^\circ &= \frac{20+h}{20}\\ \sqrt{3} &= \frac{20+h}{20}\\ 20\sqrt{3} &= 20+h\\ \implies h &= 20(\sqrt{3}-1)\end{aligned}$$

Some Applications of Trigonometry — Heights and Distances

18. 18. A vertical tower stands on a horizontal plane and is surmounted by a vertical flagstaff of height h. At a point on the plane, the angles of elevation of the bottom and the top of the flagstaff are α and β respectively. Prove that the height of the tower is $\frac{h\tan\alpha}{\tan\beta-\tan\alpha}$.

Answer:

$$\text{Height of tower} = \frac{h\tan\alpha}{\tan\beta - \tan\alpha}$$

Solution:

1. Let the height of the tower be x and the distance of the point from the base be d.
2. In the triangle formed by the tower base, the point, and the ground, $\tan\alpha = \frac{x}{d} \implies d = \frac{x}{\tan\alpha}$.
3. In the triangle formed by the flagstaff top, the point, and the ground, the total height is $x + h$.
4. $\tan\beta = \frac{x+h}{d} \implies d = \frac{x+h}{\tan\beta}$.
5. Equate the two expressions for d: $\frac{x}{\tan\alpha} = \frac{x+h}{\tan\beta}$.
6. Solve for x by cross-multiplying and rearranging terms.

Derivation:

$$\begin{aligned}
d = \frac{x}{\tan\alpha} &= \frac{x+h}{\tan\beta} \\
x\tan\beta &= (x+h)\tan\alpha \\
x\tan\beta - x\tan\alpha &= h\tan\alpha \\
x(\tan\beta - \tan\alpha) &= h\tan\alpha \\
x &= \frac{h\tan\alpha}{\tan\beta - \tan\alpha}
\end{aligned}$$

Some Applications of Trigonometry — Angle of elevation

19. 19. From a point on the ground, the angles of elevation of the bottom and the top of a transmission tower fixed at the top of a 20 m high building are 45° and 60° respectively. Find the height of the tower.

Answer:

$$20(\sqrt{3} - 1) \text{ m}$$

Solution:

1. Let h be the height of the tower and x be the distance from the point to the building.
2. Use $\tan 45^\circ = \frac{20}{x}$ to find x.
3. Use $\tan 60^\circ = \frac{20+h}{x}$ to relate h and x.
4. Substitute $x = 20$ into the second equation.
5. Solve for h and rationalize if necessary.

Derivation:

$$\begin{aligned}
\tan 45^\circ &= \frac{20}{x} \\
\implies 1 &= \frac{20}{x} \\
\implies x &= 20 \\
\tan 60^\circ &= \frac{20+h}{20} \\
\implies \sqrt{3} &= \frac{20+h}{20} \\
20\sqrt{3} &= 20 + h \\
h &= 20(\sqrt{3} - 1) \text{ m}
\end{aligned}$$

20. **20.** The angle of elevation of a cloud from a point 60 m above a lake is $30°$ and the angle of depression of the reflection of the cloud in the lake is $60°$. Find the height of the cloud from the surface of the lake.

Answer:

$$120 \text{ m}$$

Solution:

1. Let H be the height of the cloud above the lake surface. The distance of the reflection below the lake surface is also H.
2. The height of the cloud above the observation point is $H - 60$.
3. The depth of the reflection below the observation point is $H + 60$.
4. Use $\tan 30° = \frac{H-60}{d}$ and $\tan 60° = \frac{H+60}{d}$.
5. Divide the two equations to eliminate d and solve for H.

Derivation:

$$\begin{aligned} \frac{1}{\sqrt{3}} &= \frac{H-60}{d} \\ \implies d &= \sqrt{3}(H-60) \\ \sqrt{3} &= \frac{H+60}{d} \\ \implies d &= \frac{H+60}{\sqrt{3}} \\ \sqrt{3}(H-60) &= \frac{H+60}{\sqrt{3}} \\ 3(H-60) &= H+60 \\ 3H-180 &= H+60 \\ 2H &= 240 \\ \implies H &= 120 \text{ m} \end{aligned}$$

21. **21.** Two poles of equal heights are standing opposite each other side by side and opposite each other on either side of the road, which is 80 m wide. From a point between them on the road, the angles of elevation of the top of the poles are $60°$ and $30°$ respectively. Find the height of the poles and the distances of the point from the poles.

Answer:

$$\text{Height} = 20\sqrt{3} \text{ m}, \text{Distances} = 20 \text{ m and } 60 \text{ m}$$

Solution:

1. Let h be the height of both poles and x be the distance from one pole.

2. The distance from the other pole is $80 - x$.
3. Use $\tan 60° = \frac{h}{x}$ and $\tan 30° = \frac{h}{80-x}$.
4. Express h in terms of x in both equations.
5. Equate and solve for x, then find h.

Derivation:

$$\begin{aligned} h &= x\sqrt{3} \\ h &= \frac{80 - x}{\sqrt{3}} \\ x\sqrt{3} &= \frac{80 - x}{\sqrt{3}} \\ 3x &= 80 - x \\ 4x &= 80 \\ \implies x &= 20 \\ h &= 20\sqrt{3} \text{ m} \end{aligned}$$

Some Applications of Trigonometry — Two poles problem

22. **22.** If the angle between two tangents drawn from a point P to a circle of radius r and center O is $60°$, then the length of OP is:

Answer: (B)

Solution:

1. The line segment joining the center to the external point bisects the angle between the tangents.
2. The triangle formed by the center, the point of tangency, and the external point is a right-angled triangle.
3. Using trigonometry, $\sin(30°) = r/OP$.

Derivation:

$$\begin{aligned} \sin(30°) &= \frac{r}{OP} \\ \implies \frac{1}{2} &= \frac{r}{OP} \\ \implies OP &= 2r \end{aligned}$$

Circles — Tangents to a circle

23. **23.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. The length of PQ is:

Answer: (C)

Solution:

1. The tangent at any point of a circle is perpendicular to the radius through the point of contact.

2. This forms a right-angled triangle OPQ where OQ is the hypotenuse.
3. Apply the Pythagorean theorem: $PQ^2 = OQ^2 - OP^2$.

Derivation:

$$PQ = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12 \text{ cm}$$

Circles — Tangents to a circle

24. 24. The number of tangents that can be drawn from a point lying inside a circle is:

Answer: (A)

Solution:

1. A tangent touches the circle at exactly one point.
2. If a point is inside the circle, any line passing through it will intersect the circle at two points (a secant).
3. Therefore, no tangent can be drawn from an interior point.

Derivation:

$$\text{Number of tangents} = 0$$

Circles — Tangents to a circle

25. 25. If two tangents inclined at an angle of 80° are drawn to a circle of radius 3 cm, then the angle between the two radii at the center is:

Answer: (C)

Solution:

1. The quadrilateral formed by the center, two points of contact, and the external point has angles summing to 360°.
2. The angles at the points of contact are 90° each.
3. The angle at the center and the angle between tangents are supplementary.

Derivation:

$$\text{Angle} = 180^\circ - 80^\circ = 100^\circ$$

Circles — Tangents to a circle

26. 26. The length of a tangent from a point A at a distance of 10 cm from the center of the circle is 8 cm. The radius of the circle is:

Answer: (B)

Solution:

1. The radius, tangent, and the line to the external point form a right triangle.
2. The distance from the center is the hypotenuse (10), and the tangent is one leg (8).

3. Use $r^2 + 8^2 = 10^2$.

Derivation:

$$\begin{aligned} r^2 &= 100 - 64 = 36 \\ &\implies r = 6 \text{ cm} \end{aligned}$$

Circles — Tangents to a circle

27. **27.** Assertion (A): The length of a tangent drawn from an external point to a circle is equal to the length of another tangent drawn from the same point. Reason (R): The lengths of tangents drawn from an external point to a circle are equal.

Answer: (A)

Solution:

1. Identify the theorem: The lengths of tangents drawn from an external point to a circle are equal.
2. Check Assertion: It states the definition of the theorem.
3. Check Reason: It states the theorem itself.
4. Conclusion: R explains A correctly.

Derivation:

$$\begin{aligned} A &: \text{True}, R : \text{True}, R \\ &\implies A \end{aligned}$$

Circles — Tangents to a circle

28. **28.** Assertion (A): The tangent at any point of a circle is perpendicular to the radius through the point of contact. Reason (R): The perpendicular distance from the center of the circle to the tangent is equal to the radius.

Answer: (A)

Solution:

1. Recall NCERT Theorem 10.1: The tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Verify Reason: The shortest distance from the center to the tangent is the radius, identifying the perpendicularity.
3. Conclusion: Both are true and R explains A.

Derivation:

$$\text{Radius} \perp \text{Tangent at point of contact}$$

Circles — Tangents to a circle

29. **29.** Assertion (A): The number of tangents that can be drawn from a point inside a circle is zero. Reason (R): A point inside a circle lies in the interior, and any line passing through it will be a secant, not a tangent.

Answer: (A)

Solution:

1. Analyze A: A point inside the circle does not permit any tangent as a tangent must touch the circle at only one point.
2. Analyze R: Any line through an interior point intersects the circle at two points (secant).
3. Conclusion: Both true and R explains A.

Derivation:

$$\text{No tangent from interior point.}$$

Circles — Tangents to a circle

30. **30.** Assertion (A): If the angle between two tangents drawn from a point P to a circle of radius r and center O is $60°$, then $OP = 2r$. Reason (R): The line segment from the center to the external point bisects the angle between the tangents.

Answer: (A)

Solution:

1. In $\triangle OAP$ (where A is point of contact), $\angle OPA = 30°$ (bisected angle).
2. $\sin(30°) = \frac{OA}{OP} = \frac{r}{OP}$.
3. $\frac{1}{2} = \frac{r}{OP} \implies OP = 2r$.
4. Conclusion: Both are true and R explains A.

Derivation:

$$\sin(30°) = \frac{r}{OP}$$
$$\implies OP = 2r$$

Circles — Tangents to a circle

31. **31.** Assertion (A): The length of the tangent from a point A at a distance of 5 cm from the center of the circle is 4 cm. The radius of the circle is 3 cm. Reason (R): In a right triangle formed by radius, tangent, and hypotenuse (distance to center), $r^2 + t^2 = d^2$.

Answer: (A)

Solution:

1. Use Pythagoras Theorem: $r^2 + t^2 = d^2$.
2. $r^2 + 4^2 = 5^2$.
3. $r^2 + 16 = 25 \implies r^2 = 9 \implies r = 3$.

4. Conclusion: Both are true and R explains A.

Derivation:

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$

Circles — Tangents to a circle

32. **32.** A tangent PQ at a point P of a circle of radius 5 cm meets a line through the center O at a point Q so that $OQ = 13$ cm. Find the length of PQ.

Answer:

$$12cm$$

Solution:

1. Use the property that the tangent at any point of a circle is perpendicular to the radius through the point of contact.
2. Apply the Pythagorean theorem in the right-angled triangle OPQ where $\angle OPQ = 90°$.

Derivation:

$$\begin{aligned} OP^2 + PQ^2 &= OQ^2 \\ \implies 5^2 + PQ^2 &= 13^2 \\ \implies 25 + PQ^2 &= 169 \\ \implies PQ^2 &= 144 \\ \implies PQ &= 12cm \end{aligned}$$

Circles — Tangents to a circle

33. **33.** If tangents PA and PB from a point P to a circle with center O are inclined to each other at an angle of $80°$, then find $\angle POA$.

Answer:

$$50°$$

Solution:

1. In quadrilateral OAPB, $\angle OAP = \angle OBP = 90°$. The sum of angles is $360°$.
2. Find $\angle AOB = 180° - 80° = 100°$. Since $\triangle OAP \cong \triangle OBP$, $\angle POA = \frac{1}{2}\angle AOB$.

Derivation:

$$\angle AOB = 180° - 80° = 100°.\text{Thus}, \angle POA = \frac{100°}{2} = 50°$$

Circles — Tangents from an external point

34. **34.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Let O be the center, and AB be the chord of the larger circle tangent to the smaller circle at M. $OM \perp AB$.
2. In $\triangle OMA$, $OA = 5$ and $OM = 3$. Use Pythagoras to find AM, then $AB = 2AM$.

Derivation:

$$AM = \sqrt{OA^2 - OM^2} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = 4. AB = 2 \times 4 = 8cm.$$

Circles — Tangents to a circle

35. **35.** A quadrilateral ABCD is drawn to circumscribe a circle. If $AB = 6$ cm, $BC = 7$ cm, and $CD = 4$ cm, find the length of AD.

Answer:

$$3cm$$

Solution:

1. Use the property that sums of opposite sides of a quadrilateral circumscribing a circle are equal.
2. Apply $AB + CD = BC + AD$.

Derivation:

$$\begin{aligned} 6 + 4 &= 7 + AD \\ \implies 10 &= 7 + AD \\ \implies AD &= 3cm. \end{aligned}$$

Circles — Tangents to a circle

36. **36.** If a point P is 26 cm away from the center O of a circle and the length of the tangent PT drawn from P to the circle is 10 cm, find the radius of the circle.

Answer:

$$24cm$$

Solution:

1. In $\triangle OTP$, $\angle OTP = 90°$. OT is the radius.
2. Use Pythagoras: $OT^2 + PT^2 = OP^2$.

Derivation:

$$\begin{aligned} OT^2 + 10^2 &= 26^2 \\ \implies OT^2 + 100 &= 676 \\ \implies OT^2 &= 576 \\ \implies OT &= 24cm. \end{aligned}$$

Circles — Tangents to a circle

37. **37.** A quadrilateral ABCD is drawn to circumscribe a circle. Prove that $AB+CD = AD + BC$.

Answer:

$$AB + CD = AD + BC$$

Solution:

1. Use the property that lengths of tangents drawn from an external point to a circle are equal.
2. Write equations for tangents from vertices A, B, C, and D.
3. Add the equations and group the terms to form the sides of the quadrilateral.

Derivation: Let the circle touch sides AB, BC, CD, DA at P, Q, R, S respectively. AP=AS, BP=BQ, CR=CQ, DR=DS. Adding them: (AP+BP)+(CR+DR) = (AS+DS)+(BQ+CQ). Thus, AB+CD = AD+BC. *Circles — Tangents to a circle*

38. **38.** From a point Q, the length of the tangent to a circle is 24 cm and the distance of Q from the centre is 25 cm. Find the radius of the circle.

Answer:

$$7cm$$

Solution:

1. Identify that the radius is perpendicular to the tangent at the point of contact.
2. This forms a right-angled triangle with the radius, tangent, and distance from the center as sides.
3. Apply the Pythagorean theorem to solve for the radius.

Derivation:

$$\begin{aligned} r^2 + 24^2 &= 25^2 \\ \Rightarrow r^2 + 576 &= 625 \\ \Rightarrow r^2 &= 49 \\ \Rightarrow r &= 7 \text{ cm}. \end{aligned}$$

Circles — Tangents to a circle

39. **39.** Two concentric circles are of radii 5 cm and 3 cm. Find the length of the chord of the larger circle which touches the smaller circle.

Answer:

$$8cm$$

Solution:

1. Draw a radius to the point of contact on the smaller circle, which acts as the perpendicular bisector of the chord.
2. Create a right triangle using the radius of the small circle, half the chord, and the radius of the large circle as the hypotenuse.
3. Calculate half the chord length and multiply by 2.

Derivation:

$$\text{Half}-\ \text{chord} = \sqrt{5^2 - 3^2} = \sqrt{25 - 9} = \sqrt{16} = 4\text{ cm.Total length} = 2 \times 4 = 8\text{ cm.}$$

Circles — Tangents to a circle

40. **40.** If tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°, then find $\angle POA$.

Answer:

$$50^\circ$$

Solution:

1. Recognize that the line from the center to the external point bisects the angle between the tangents.
2. Consider the quadrilateral OAPB where angles at A and B are 90°.
3. Determine the angle $\angle APB$ is split in half and use the sum of angles in triangle OAP.

Derivation:

$$\angle APB = 80^\circ$$
$$\Rightarrow \angle OPA = 40^\circ.\text{In } \triangle OAP, \angle OAP = 90^\circ.\angle POA = 180^\circ - 90^\circ - 40^\circ = 50^\circ.$$

Circles — Tangents to a circle